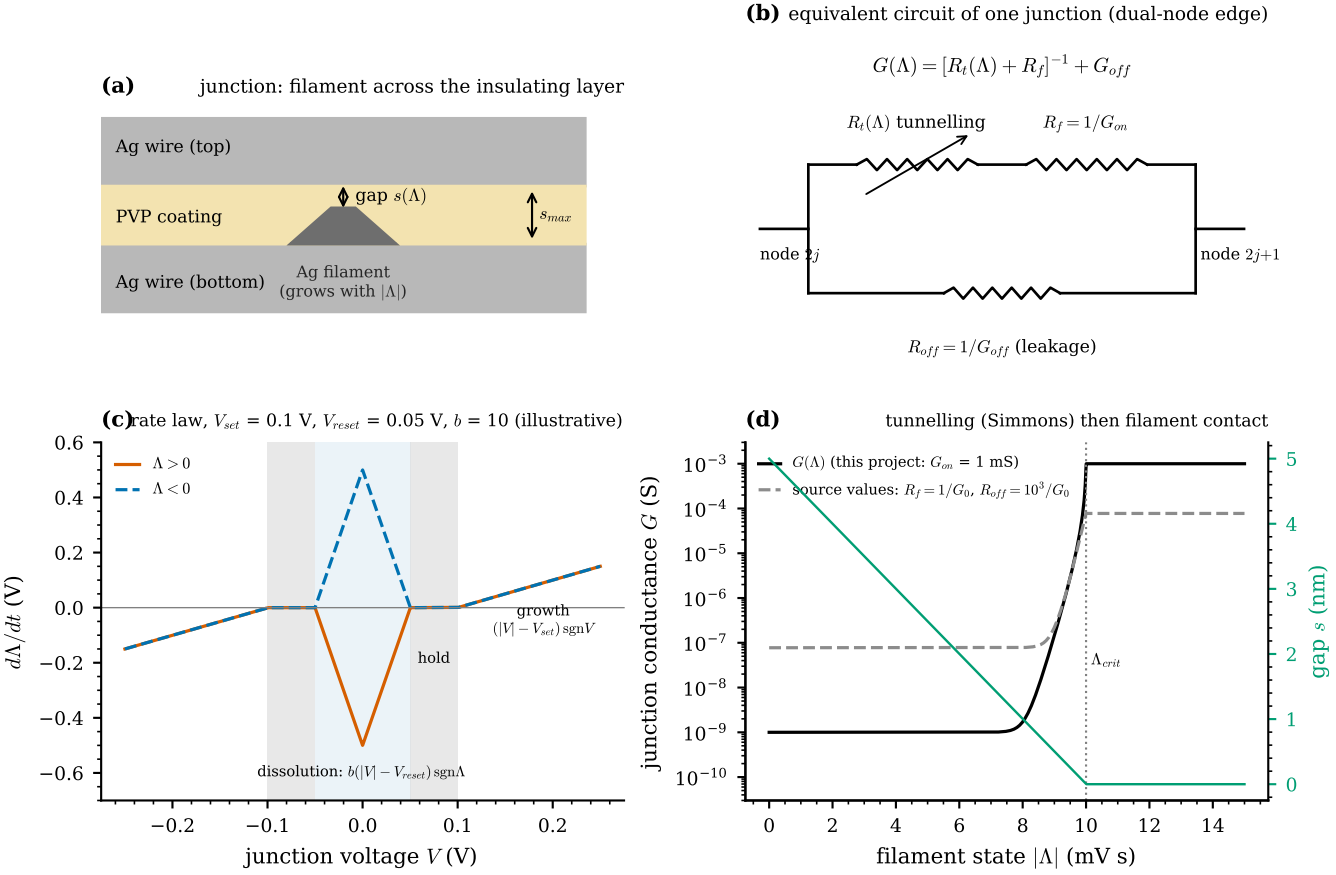
(a) junction: filament across the insulating layer
Ag wire (top)
PVP coating
gap $s(\Lambda)$
s_{max}
Ag wire (bottom)
Ag filament
(grows with $|\Lambda|$)
(b) equivalent circuit of one junction (dual-node edge)
$G(\Lambda) = [R_t(\Lambda) + R_f]^{-1} + G_{off}$
$R_t(\Lambda)$ tunnelling
$R_f = 1/G_{on}$
node $2j$
node $2j+1$
$R_{off} = 1/G_{off}$ (leakage)
(c) rate law, V_{set} = 0.1 V, V_{reset} = 0.05 V, b = 10 (illustrative)
$\Lambda > 0$
$\Lambda < 0$
growth
$(|V| - V_{set})\,\mathrm{sgn}V$
hold
dissolution: $b(|V| - V_{reset})\,\mathrm{sgn}\Lambda$
$d\Lambda/dt$ (V)
junction voltage V (V)
(d) tunnelling (Simmons) then filament contact
$G(\Lambda)$ (this project: G_{on} = 1 mS)
source values: $R_f = 1/G_0$, $R_{off} = 10^3/G_0$
Λ_{crit}
junction conductance G (S)
gap s (nm)
filament state $|\Lambda|$ (mV s)